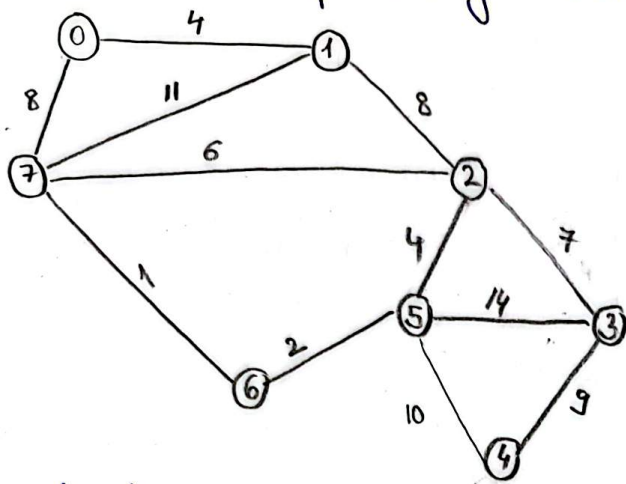


Minimum Spanning Tree (Prim)

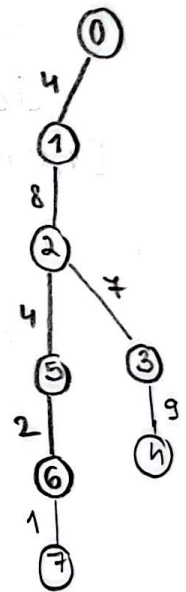


Edges sorted:

$(6, 7) : 1$	$(0, 4) : 8$
$(6, 5) : 2$	$(1, 2) : 8$
$(0, 1) : 4$	$(3, 4) : 9$
$(2, 5) : 4$	$(5, 4) : 10$
$(2, 7) : 6$	$(1, 7) : 11$
$(2, 3) : 7$	$(3, 5) : 14$

start : node 0

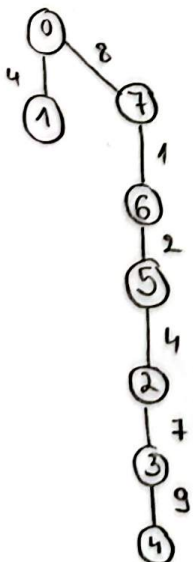
	pop from pq	visited nodes	(c, m, p) pq (sorted)
step 1	-	0	$[(4, 1, 0), (8, 7, 0)]$
step 2	$(4, 1, 0)$	0, 1	$[(4, 0, 1), (8, 7, 0), (8, 2, 1), (11, 7, 1)]$
step 3	$(4, 0, 1)$	skip	$[(8, 2, 1), (8, 7, 0), (11, 7, 1)]$
step 4	$(8, 2, 1)$	0, 1, 2	$[(4, 5, 2), (6, 7, 2), (7, 3, 2), (8, 7, 0), (8, 1, 2), (11, 7, 1)]$
step 5	$(4, 5, 2)$	0, 1, 2, 5	$[(2, 6, 5), (6, 7, 2), (7, 3, 2), (8, 7, 0), (8, 1, 2), (11, 7, 1), (4, 2, 5), (14, 3, 5), (10, 4, 5)]$
step 6	$(2, 6, 5)$	0, 1, 2, 5, 6	$[(1, 7, 6), (6, 7, 2), (7, 3, 2), (8, 7, 0), (8, 1, 2), (11, 7, 1), (4, 2, 5), (14, 3, 5), (10, 4, 5), (2, 5, 6)]$
step 7	$(1, 7, 6)$	0, 1, 2, 5, 6, 7	$[(6, 2, 7), (7, 3, 2), (8, 0, 7), (10, 4, 5) \dots]$
step 8	$(6, 2, 7)$	skip (2 was visited)	
step 9	$(7, 3, 2)$	0, 1, 2, 3, 5, 6, 7	$[(9, 4, 3), (10, 4, 5), (14, 5, 3)]$
step 10	$(9, 4, 3)$	0, 1, 2, 3, 4, 5, 6, 7	$[(10, 5, 4)]$

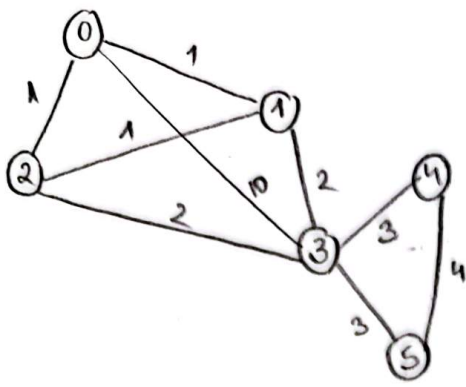


total cost:

$$4 + 8 + 4 + 2 + 1 + 7 + 9 = 35$$

Another MST with the same cost:



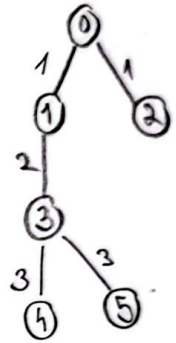


Edges sorted:

$(0,1):1$ $(3,4):3$
 $(0,2):1$ $(3,5):3$
 $(1,2):1$ $(4,5):4$
 $(1,3):2$ $(0,3):10$
 $(2,3):2$

start : node 0

	pop from pq	visited nodes	(c, m, p) pq(sorted)
step 1	-	0's	$[(1,1,0), (1,2,0), (10,3,0)]$
step 2	$(1,1,0)$	0,1's	$[(1,2,0), (1,2,1), (2,3,1), (10,3,0)]$
step 3	$(1,2,0)$	0,1,2's	$[(1,2,1), (2,3,1), (2,3,2), (10,3,0)]$
step 4	$(1,2,1)$	skip (2 was visited)	
step 5	$(2,3,1)$	0,1,2,3's	$[(2,3,2), (3,4,3), (3,5,3), (10,3,0)]$
step 6	$(2,3,2)$	skip (3 was visited)	
step 7	$(3,4,3)$	0,1,2,3,4's	$[(3,5,3), (4,5,4), (10,3,0)]$
step 8	$(3,5,3)$	0,1,2,3,4,5's	$[(4,5,4), (10,3,0)]$



total cost:

$$1 + 2 + 3 + 3 + 1 = 10$$

All MST with total cost = 10

